

AI(CL-461)

Date: February, 9th 2026

Course Instructor(s)

Mr. Usman Anwer



Quiz-1

Total Time: 20 mins

Total Marks: 20

Total Questions: 01

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Student Name	Roll No	Section	Student Signature

Instructions: Apply DFS by demonstrating states of open and closed lists step by step.

CLO 3: Apply methods of blind as well as informed search on the problem related to the programs.

Question 1:

There are three missionaries and three cannibals(Dracula's) on one side of a river.

1. A boat is available that can carry at most two people, excluding the person operating (rowing) the boat.
2. At no time should the number of cannibals exceed the number of missionaries on either river bank, provided missionaries are present.
3. The objective is to safely transport all missionaries and cannibals to the opposite bank.
4. The boat must be operated by at least one person and can travel back and forth across the river. This formulation specifies the initial state, goal state, and constraints of the problem.

Initial state is: **(3, 3, L)** and goal state is: **(0,0,R)** while a state is represented as (M_left, C_left, Boat)
(M_left=> No of missionaries on left , no of cannibals on left and Boat $\in \{L, R\}$).

Possible boat action moves are:

Move	Description
(2M)	Two missionaries
(2C)	Two cannibals
(1M,1C)	One missionary & one cannibal
(1M)	One missionary
(1C)	One cannibal

First task is to expand state space and then APPLY DFS over it to find goal state from start.